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Lecturer : Nurul Amara binti Muhamad Nazmi

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Date Received :

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Shielim Keisha Nalani

SWE2309441

Date: 23/04/2026

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1. App Design Concept & UI Alignment

1.1. App Concept, Purpose, and Target Users

HavaPaw is a mobile application for a smart pet collar system, which is an IoT wearable device worn by cats and dogs. The app is a user interface for all the collar data, which transforms the sensor readings into insights for the pet owners.

The problem HavaPaw addresses is a common frustration among pet owners. They cannot monitor their pet's whereabouts and health in real time, which may have led to injuries, critical illnesses or escape. The existing solutions are either too expensive or too complex. Meanwhile, HavaPaw is able to resolve those issues in a single mobile dashboard.

The app's purpose is to provide real-time physical location of the pet via GPS, display health metrics such as heart rate, body temperature, activity levels and calorie burnt, as well as alerting the owner when the pet leaves a specific safe zone or shows an abnormal pattern, such as an extremely high heart rate or sudden inactivity.

The primary target users for this mobile application are dog and cat users aged 18-45 who are health-conscious about their pets and regularly spend time away from home, whether at work, travelling, or on occasions. These users are supposedly comfortable with using smartphone apps. The secondary target users are elderly pet owners who are less aware of digital systems, and first-time pet owners who need guidance.

1.2. UI Alignment with Modern Mobile Design Trends

HavaPaw's UI is designed based on several mobile design trends. Firstly, the minimalist layout where each screen has a single primary action. It improves the readability, directs attention to important actions, and creates a modern look (Smith, 2025). The dashboard doesn't show everything at once, instead it only displays the most important metrics and the map preview.

The app also follows the bottom tab navigation pattern. HavaPaw follows a four-tab bar with Home, Health, Map and Profile. Placing navigation at the bottom enables users to move comfortably with one hand. In addition, it's also a standard navigation panel that is commonly

used for its accessibility and appeal (Caposino, 2025). The active tab is highlighted in a teal colour to show the user's current location within the app.

Another design trend is the use of semantic colour. Rather than applying colour for only decorative purposes, HavaPaw uses teal for normal and healthy state, amber for warning conditions, and red for critical alerts (Interaction Design Foundation, 2021). This allows mental understanding, so the users can access their pet's status at a glance without reading any text. Applying the same colour rules consistently across every screen builds familiarity towards the HavaPaw brand identity (Henrique, 2025).

The health screen will show data visualization through bar and line charts. For example, showing heart rates hourly for the last 12 hours helps identify when a spike occurred and how long it lasted. When apps handle complex data, presenting the information visually rather than as raw numbers help users interpret data quickly on small screens (Smith, 2025). Finally, the overall interface uses a card-based layout, making the UI more organized. It makes the interface easy to navigate on any screen size, while preventing the layout from feeling too cluttered (Magic UI, 2025).

1.3. Design Decisions

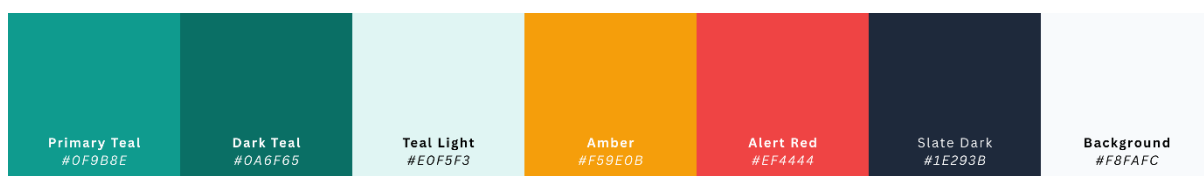


Figure 1.1. Colour Palette

The colour palettes in Figure 1.1 are all chosen to support the purpose of the app. Teal (Green) symbolises nature, growth, freshness, and quality, which is appropriate for a health monitoring application. Meanwhile, amber is used to signal warnings where something needs attention. Red is then reserved for critical alerts (Interaction Design Foundation, 2021). The three teal tones create a layered visual that gives a consistent look without being flat. In addition, the black and white colours are the basic colour that helps create a modern look for the application.

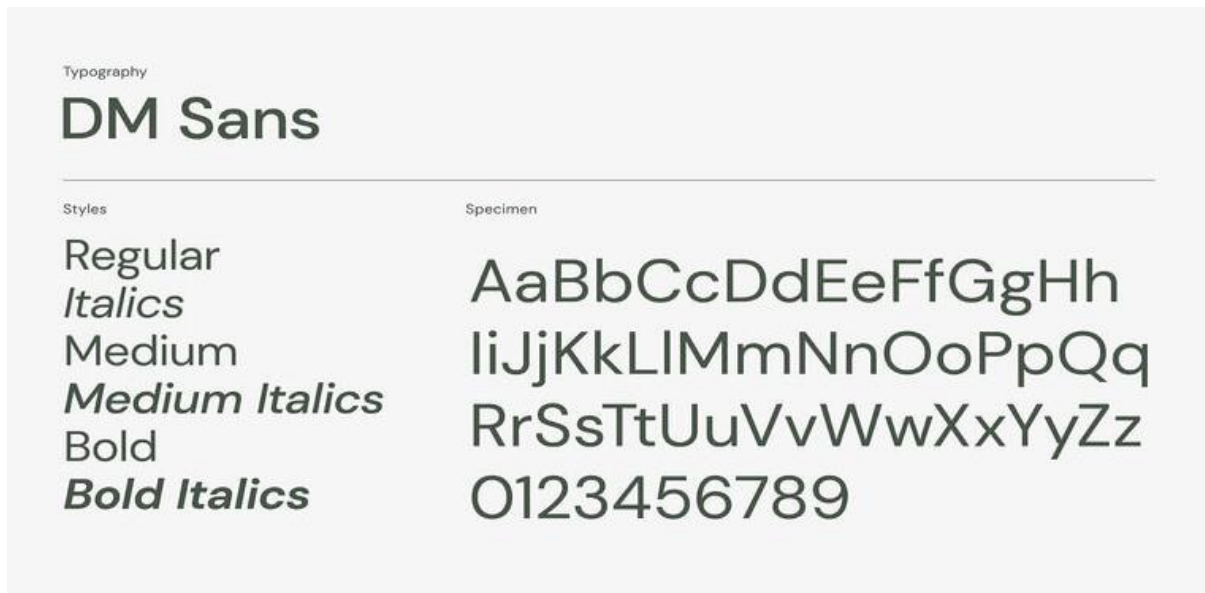


Figure 1.2. DM Sans Text Typeface Font (Softlay, 2025)

For the typeface, DM Sans, shown in Figure 1.2, is used as it looks cute with the rounded letterforms, which fits the pet care theme. In addition, the letters are clear and basic, which makes the visual more user friendly, especially since the text should be more readable on small mobile screens.

All buttons, cards, and input fields in HavaPaw use rounded corners rather than sharp edges. This is intentional since many users think rounded shapes look friendlier and more accessible than angular ones. It's an important quality for a consumer pet care product. Rounded corners also improve the visuals by softening the boundaries between elements, especially on small screens (Mukherjee, 2023).

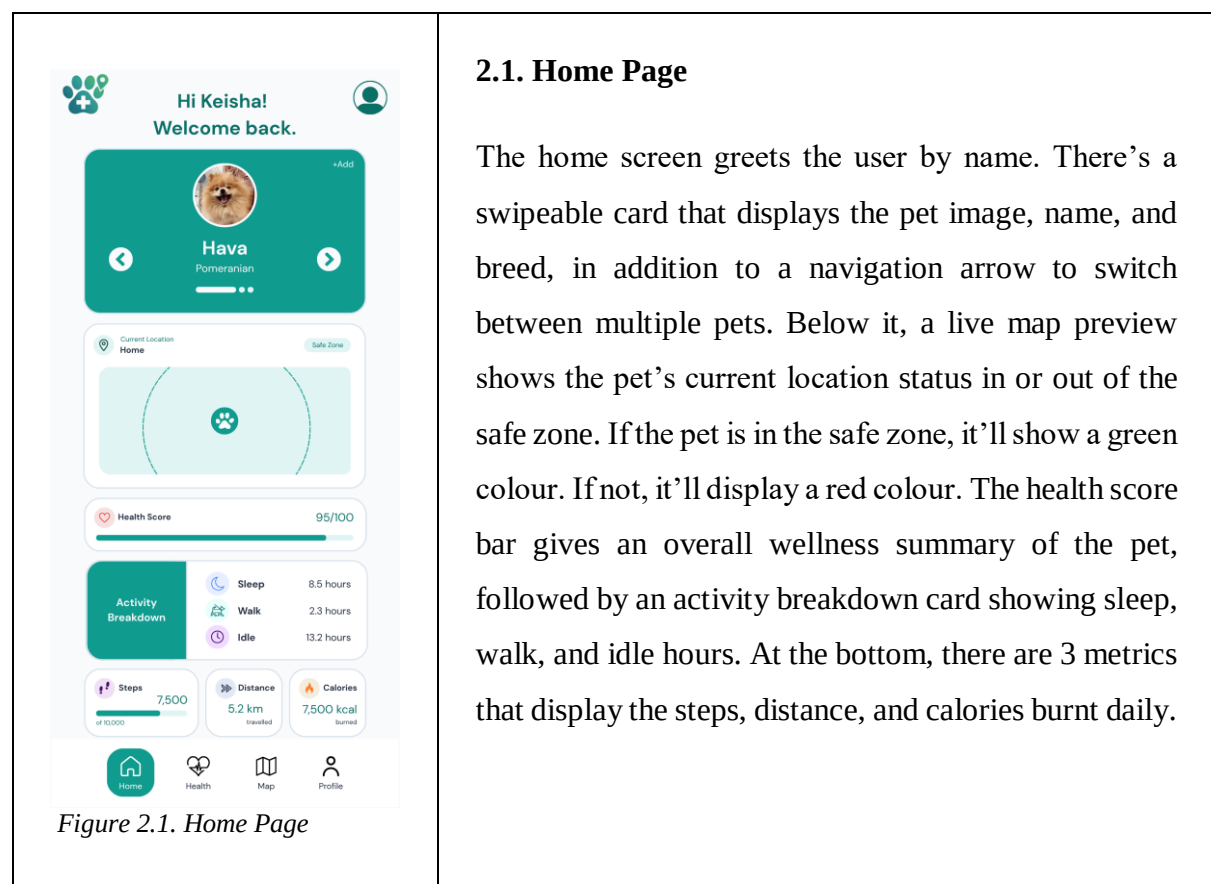
A mini map preview is displayed on the home dashboard rather than only showing it on the Map tab to allow quick checks for the pet owners. Most of the time, pet owners would be curious on "Where is my pet right now?". When opening the app, it'll be easier for them to view it right away as it's located in the home tab. Additionally, the notifications of alerts are not displayed inside a notification tab, instead they will be shown on the health tab itself for the users who'd want to know more about their pet's health status.

The pet details on the profile screen is also a design choice. The user interface on the profile tab must be centred around the pet's picture and name. Instead of making the user look at the profile tab of his/her account, HavaPaw shows their pet profiles and details there. Their own

account details can be viewed and modified in the Account Profile in the top right corner of the screen.

Finally, the name HavaPaw carries a personal meaning. Hava is my first dog (out of three), and this app is designed as a dedication to them. The "Paw" in the name connects to its purpose as a pet monitoring application. Rather than choosing a generic product name, naming the app after a real pet gives it a more emotional feel and motivation behind the project.

2. Mobile Application Overall Structure



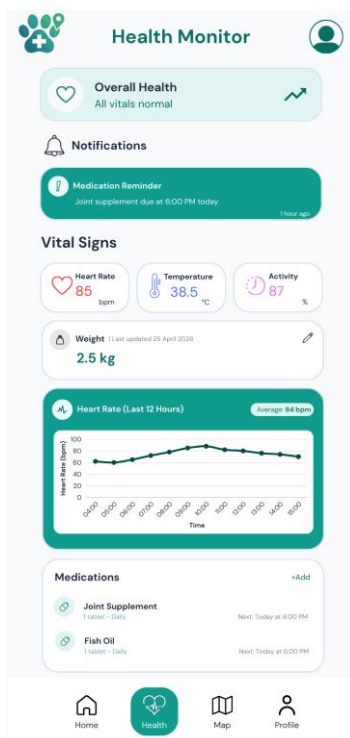


Figure 2.2. Health Page

2.2. Health Page

The health monitor screen provides a detailed view of the pet's physiological data. At the top, there's an overall health card that indicates the current health status. If the health status has no abnormalities, it'll be green. However, if the health status has abnormalities, it may be displayed in an amber or red colour, depending on the severity. The vital signs section displays the heart rate, temperature, and activity percentage. Below it is the weight card that shows the last recorded measurement with an edit option. There's also a heart rate line graph showing the last 12 hours of data with its average. Lastly, the medication section at the bottom contains the medication schedules with their next dosage times.

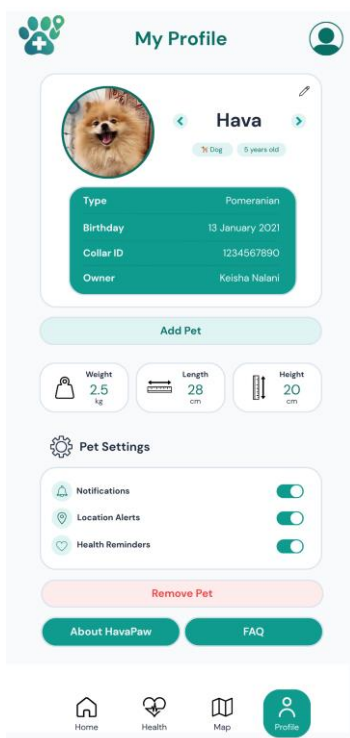


Figure 2.3. Profile Page

2.3. Profile Page

The profile screen is centred around the pet rather than the owner's account to give pet-first image. The pet's photo, name, pet type, and age are displayed at the top, also with navigation arrows to switch between pets. Below are more detailed information including breed type, collar ID, and owner name. There are also physical measurements: weight, length and height. The pet settings contain toggles for notifications, location alerts, and health reminders. In order to add or remove pet, simply press either buttons.

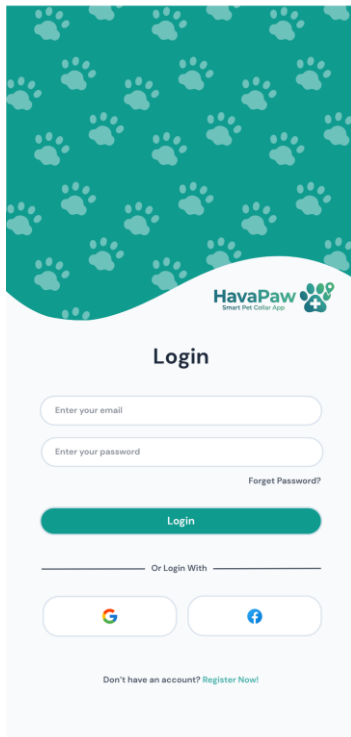


Figure 2.4. Login Page

2.4. Login Page

The login page is the entry point of HavaPaw. It shows a teal paw print pattern background, as well as the HavaPaw logo right below it. It's followed by the email and password input fields. Users can also recover their password via the "Forgot Password?" link. Once all input fields are entered, the user can simply enter the login button. There're also options to login with Google and Facebook, in addition to a link to register a new account for the new users.

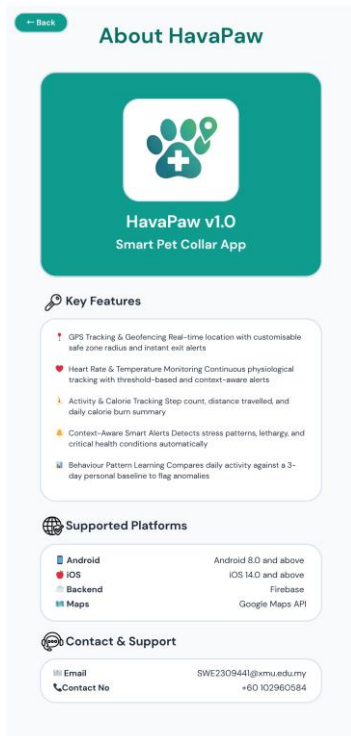


Figure 2.5. About Page

2.5. About Page

The about screen displays the app name, tagline ("Smart Pet Collar App"), and version number. The key features section shows all five core capabilities of the app. The supported platforms section shows the platform version compatible with the app. Lastly, there's a contact & support section that displays the developer's email and contact number.

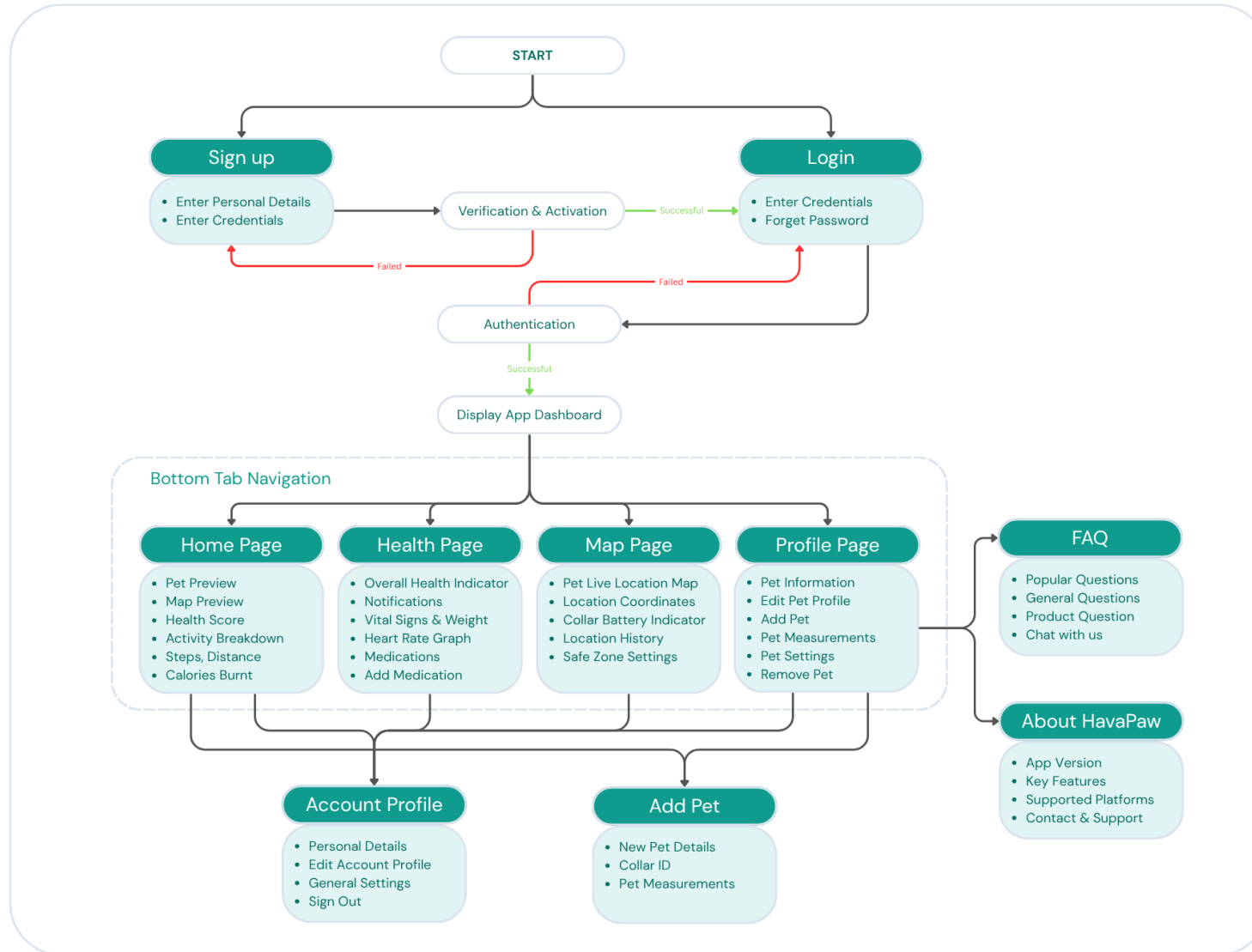


Figure 3.1. Navigation Structure Diagram

3. App Design Programming Concepts

3.1. App Navigation Structure

Figure 3.1 shows the navigation structure of the HavaPaw. The app uses a stack navigator during the authentication flow before the main app, including the sign up and login. Once authenticated, there's a bottom tab navigator that manages 4 sections which are: Home, Health, Map, and Profile. However, there are still sub-screens such as account profile, add pet, FAQ, and About HavaPaw.

3.2. Screen Transitions

HavaPaw uses several types of transitions depending on the relationship between the screens involved:

3.2.1. Authentication to Home Transition

When a user successfully logs in, the app moves them from the Login screen to the Home Dashboard. Once the user is authenticated, they cannot press back to return to the Login screen unless the account is signed out. This prevents users from accidentally leaving the main app after signing in.

3.2.2. Bottom Tab Navigation Switch Transition

Switching between the four main tabs (Home, Health, Map, and Profile) happens through the bottom navigation bar. Since all four tabs are equal in hierarchy, the screen instantly swaps. The active tab icon changes colour to teal to confirm the user's current location. The map preview on the home screen also uses this tab switch. Tapping it will switch the active tab to the Map page.

3.2.3. Stack Push Transition

When a user taps a button that opens a new separate screen, including the FAQ page, About HavaPaw, and Account Profile Page from the Profile tab. The screen being redirected to will depend on the respective buttons. For example, pressing the FAQ button will open the FAQ page with the general questions listed. A back arrow will appear on the top left corner, which allows the user to go back to where they came from.

3.2.4. Popup Overlay Transition

Several actions in HavaPaw do not open a new screen, instead a panel appears over the current screen without replacing it. This applies to Add Pet, Remove Pet, Edit Pet Profile, Edit Account Profile, and Add Medication. For several functions such as Remove Pet, a confirmation panel appears first before the action is executed, giving the user a chance to cancel. These popups can be dismissed by tapping outside them or pressing the close button, returning the user to the screen they were already on.

4. Reflection

Designing HavaPaw was an interesting experience that helped me understand what it truly takes to build a mobile application. The biggest challenge was deciding what should be prioritised based on what a pet owner actually needs. This actually shaped most of the design decisions made in this project.

I also learned that colour should work as a communication tool. Initially, I chose teal simply because it looked appealing for a health/gps app. However, as I designed the app, I thought it'd be better if the users could understand their pet's condition at a glance without reading a single word. It was a simple change but it significantly improved the usefulness of the design.

If I were to improve this in the future, I would conduct real user testing with actual pet owners, as most designs were just my assumptions as a pet owner. Another is that I would like to explore dark modes, as it's increasingly demanded in the modern apps. Finally, I'd like to add more functionalities such as machine learning to identify patterns and give insights to the pet's health. Overall, this project gave me better understanding on how to build a good mobile app UI.

References

- Caposino, M. (2024, November 29). *UI UX design trends that will transform 2026*. Fuselab Creative. <https://fuselabcreative.com/ui-ux-design-trends-2026-modern-ui-trends-ux-trends-guide/>
- Caposino, M. (2025, March 6). *20 mobile app design trends for 2026 you need to know*. Fuselab Creative. <https://fuselabcreative.com/mobile-app-design-trends-for-2025/>
- Chhay, S. (2025, August 26). *Color psychology: How veterinary practices can use it in marketing and branding*. LifeLearn Animal Health. <https://www.lifelearn.com/2025/08/26/color-psychology-how-veterinary-practices-can-use-it-in-marketing-and-branding/>
- Henrique, R. (2025, August 15). *The psychology of color in UX and digital products*. Smashing Magazine. <https://www.smashingmagazine.com/2025/08/psychology-color-ux-design-digital-products/>
- Interaction Design Foundation. (2021, November 4). *What is color symbolism?* IxDF. <https://ixdf.org/literature/topics/color-symbolism>
- Magic UI. (2025, November 1). *A guide to modern cards UI design*. Magic UI Blog. <https://magicui.design/blog/cards-ui-design>
- Mukherjee, P. (2023, February 20). *5 reasons rounded UI design is taking over digital products in 2025*. Medium. <https://medium.com/@theupbeatdesigner/exploring-the-popularity-of-rounded-ui-design-why-its-everywhere-781eea809f8>
- Smith, C. (2025, March 25). *Top mobile design trends every designer should know*. Medium. <https://medium.com/@CarlosSmith24/top-mobile-design-trends-every-designer-should-know-168f45e55bf5>
- Softlay. (2025). *DM Sans font free download*. Softlay. <https://www.softlay.com/downloads/dm-sans-font>

APPENDIX 1

MARKING RUBRICS

Criteria	Poor (0-2)	Average (3-5)	Good (6-8)	Excellent (9-10)	Weight (%)	Mark
App Concept & Creativity (Task 1)	Idea unclear, lacks relevance or originality.	Basic idea stated but limited creativity or user focus.	Clear and relevant idea with some innovative elements.	Strong, creative concept aligned with current mobile design trends and user needs.	10	
	Poor (0-2)	Average (3-5)	Good (6-8)	Excellent (9-10)		
UI Layout & Design Quality (Task 2)	Unclear layout, inconsistent color/font usage, poor spacing.	Layout is understandable but lacks balance and polish.	Organized layout with good color contrast and usability.	Professional, visually cohesive layout with excellent balance and intuitive design.	10	
	Poor (0-2)	Average (3-5)	Good (6-8)	Excellent (9-10)		
Navigation Flow & Logic (Task 3)	No or incomplete navigation flow; unclear screen linkage.	Basic flow shown but lacks logical connections or clarity.	Functional and logical navigation flow with some transitions described.	Well-structured, clearly explained navigation with effective screen relationships.	10	
	Poor (0-2)	Average (3-5)	Good (6-8)	Excellent (9-10)		
Reflection & Continuous Learning (Task 4)	No reflection or poor understanding of design improvement.	Minimal reflection with general comments only.	Clear and thoughtful reflection with self-assessment.	Insightful reflection showing awareness of learning, improvement, and user-centred design.	10	
Total						/40

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Keisha Nalani Shielim

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